

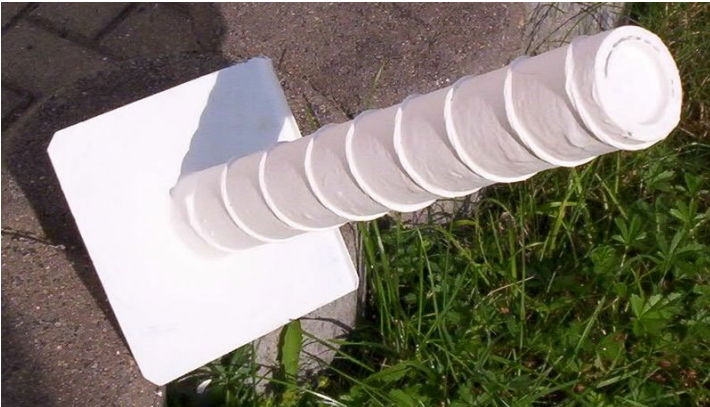
ECE 3317

Applied Electromagnetic Waves

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Fall 2025

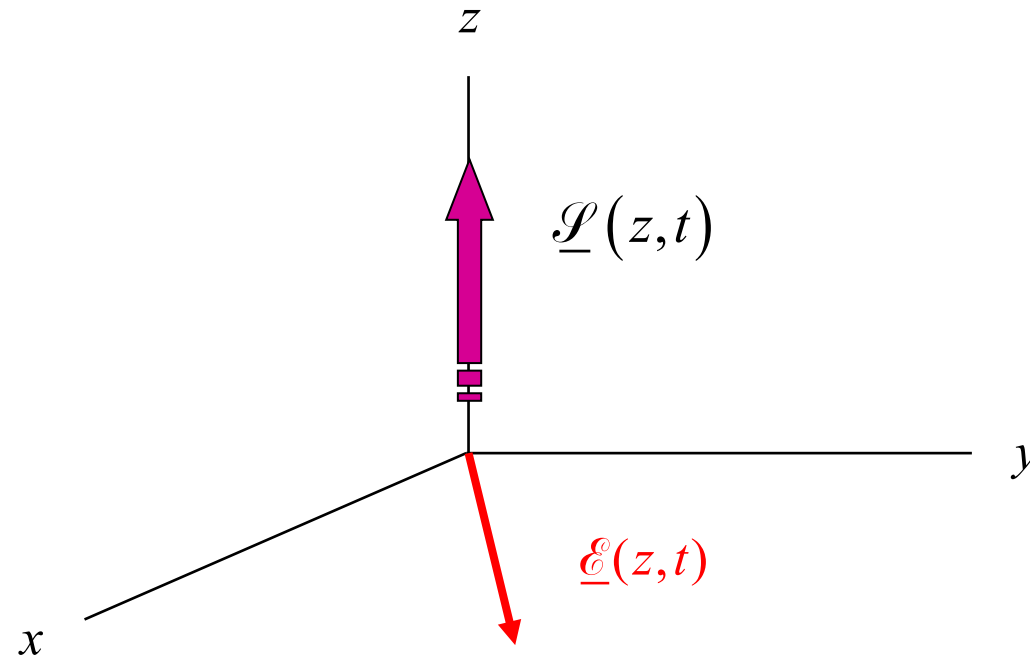
Notes 17

Polarization of Plane Waves



Polarization

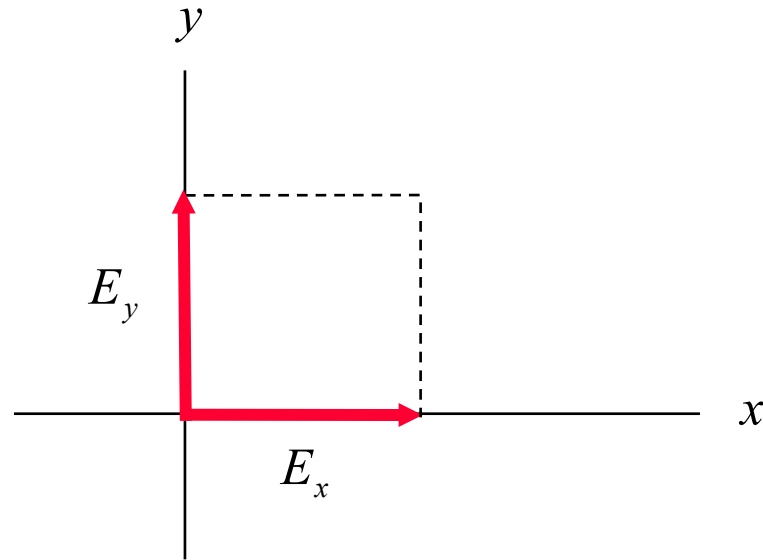
The polarization of a plane wave refers to the direction of the electric field vector in the time domain.



We assume here that the wave is traveling in the positive z direction.

Polarization (cont.)

Consider a plane wave with both x and y components



Phasor domain:

$$\underline{E}(z) = (\underline{\hat{x}} E_x + \underline{\hat{y}} E_y) e^{-jk_0 z}$$

(We assume here that the wave is traveling through free space in the z direction.)

Assume:

$$E_x = a = \text{real number}$$

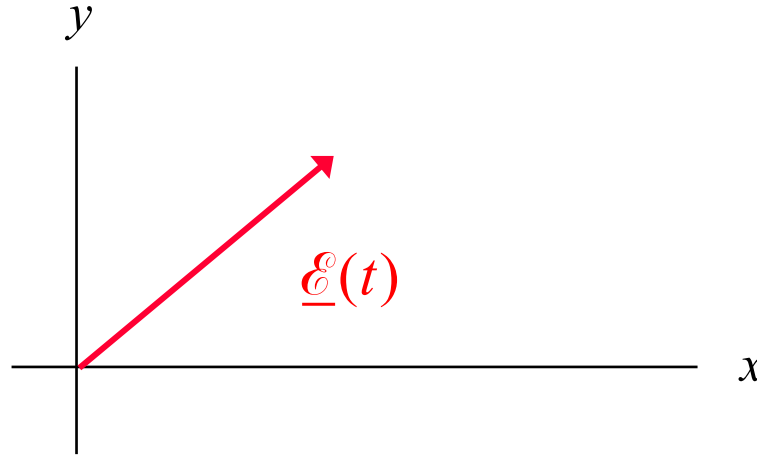
$$E_y = b e^{j\beta}$$

(In general, $\beta = \text{phase of } E_y - \text{phase of } E_x$)

Polarization (cont.)

Time Domain:

$$z = 0$$



$$\mathcal{E}_x = \operatorname{Re}\left(a e^{j\omega t}\right) = a \cos(\omega t)$$

$$\mathcal{E}_y = \operatorname{Re}\left(b e^{j\beta} e^{j\omega t}\right) = b \cos(\omega t + \beta)$$

Depending on b/a and β , three different cases arise:

- Linear polarization
- Circular polarization
- Elliptical polarization

Polarization (cont.)

Power Density:

$$\underline{S} = \frac{1}{2} \underline{E} \times \underline{H}^*$$

From Faraday's law: $H_y = \frac{E_x}{\eta_0}$, $H_x = -\frac{E_y}{\eta_0}$ $\left(\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} \doteq 376.7303 [\Omega] \right)$

Hence, we have:

$$\underline{S} = \frac{1}{2} (\hat{x} E_x + \hat{y} E_y) \times \left(\hat{x} \left(-\frac{E_y}{\eta_0} \right) + \hat{y} \left(\frac{E_x}{\eta_0} \right) \right)^*$$

For lossless free-space medium (η_0 is real):

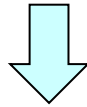
$$\underline{S} = \frac{1}{2\eta_0} \hat{z} \left(|E_x|^2 + |E_y|^2 \right) \quad \text{or} \quad \underline{S} = \frac{1}{2\eta_0} \hat{z} |\underline{E}|^2$$

Linear Polarization

$$\beta = 0 \quad \text{or} \quad \beta = \pi$$

At $z = 0$:

$$\begin{cases} \mathcal{E}_x = a \cos(\omega t) \\ \mathcal{E}_y = b \cos(\omega t + \beta) \end{cases}$$



$$\begin{cases} \mathcal{E}_x = a \cos \omega t \\ \mathcal{E}_y = \pm b \cos \omega t \end{cases} \quad \begin{cases} +\text{sign: } \beta = 0 \\ -\text{sign: } \beta = \pi \end{cases}$$

$$\underline{\mathcal{E}} = (\underline{\hat{x}} a \pm \underline{\hat{y}} b) \cos \omega t$$

Recall:

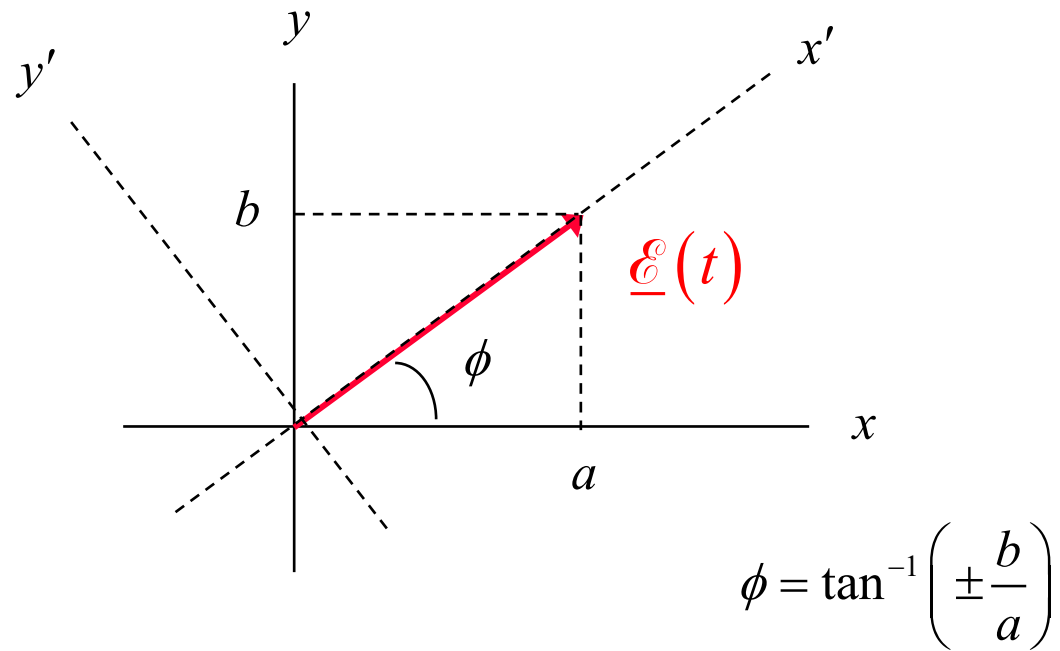
$$E_x = a = \text{real number}$$

$$E_y = b e^{j\beta}$$

Linear Polarization (cont.)

$$\underline{\mathcal{E}} = (\underline{\hat{x}} a \pm \underline{\hat{y}} b) \cos \omega t$$

This is simply a “tilted” plane wave.



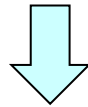
(shown for $\beta = 0$)

Circular Polarization

$$b = a \quad \text{and} \quad \beta = \pm\pi / 2$$

At $z = 0$:

$$\begin{cases} \mathcal{E}_x = a \cos(\omega t) \\ \mathcal{E}_y = b \cos(\omega t + \beta) \end{cases}$$



$$\mathcal{E}_x = a \cos \omega t$$

$$\mathcal{E}_y = a \cos(\omega t \pm \pi / 2) = \mp a \sin \omega t$$

Recall:

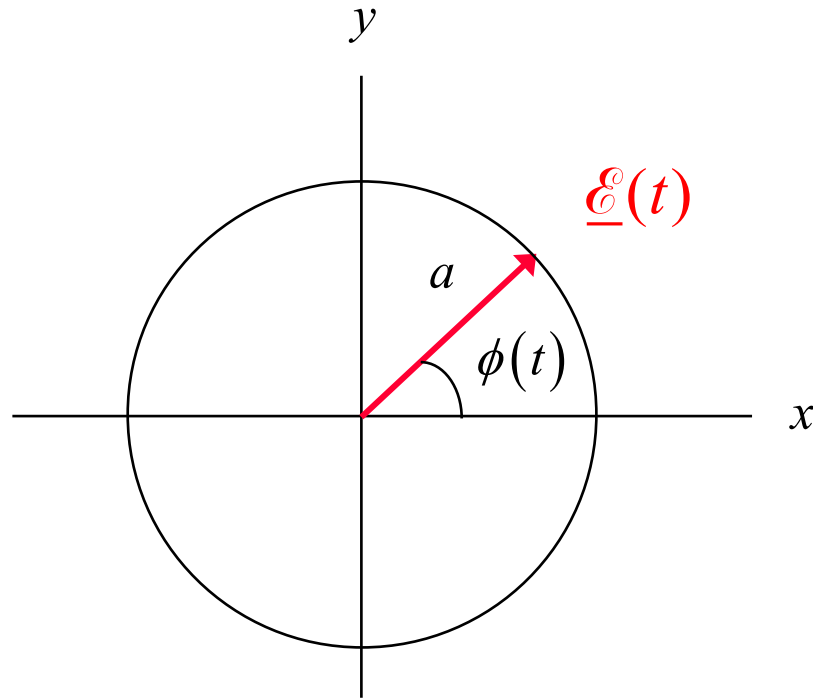
$$E_x = a = \text{real number}$$

$$E_y = b e^{j\beta}$$

Note: The top sign is always for $\beta = +\pi/2$.

$$|\underline{\mathcal{E}}|^2 = \mathcal{E}_x^2 + \mathcal{E}_y^2 = a^2 \cos^2 \omega t + a^2 \sin^2 \omega t = a^2$$

Circular Polarization (cont.)



$$\phi(t) = \tan^{-1} \left(\frac{\mathcal{E}_y}{\mathcal{E}_x} \right) = \tan^{-1} (\mp \tan \omega t) = \mp \tan^{-1} (\tan \omega t)$$

$$\Rightarrow \phi(t) = \mp \omega t \Rightarrow \frac{d\phi(t)}{dt} = \mp \omega$$

Circular Polarization (cont.)

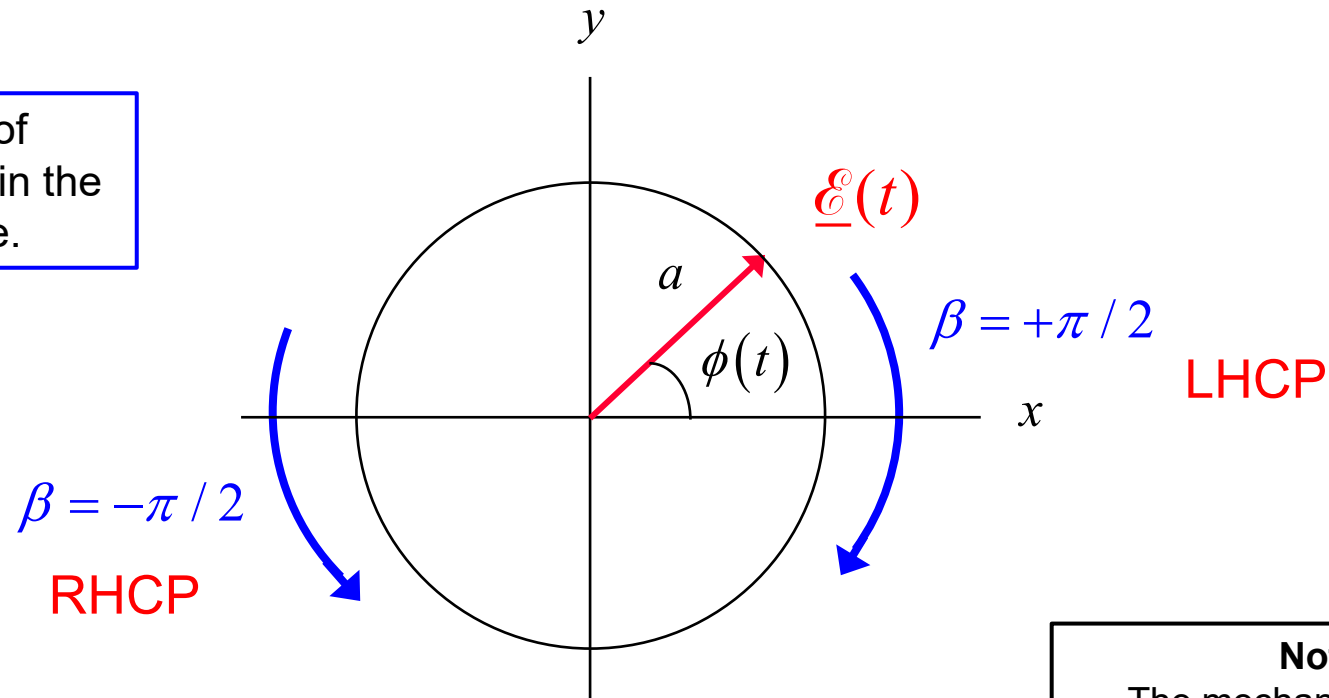
IEEE convention:

Your thumb is in the direction of propagation, and your fingers are in the direction of the rotation in time.

$$\beta = \pm \pi / 2$$

$$\mathcal{E}_x = a \cos(\omega t)$$

$$\mathcal{E}_y = b \cos(\omega t + \beta)$$



$$\frac{d\phi(t)}{dt} = \mp \omega$$

Note:
The mechanical angular velocity is the same as the electrical radian frequency ω .

Circular Polarization (cont.)

Rotation in space vs. rotation in time

Examine how the field varies in both space and time:

$$\underline{E}(z) = \left(\underline{\hat{x}} a + \underline{\hat{y}} b e^{j\beta} \right) e^{-jk_0 z} \quad \text{Phasor domain}$$



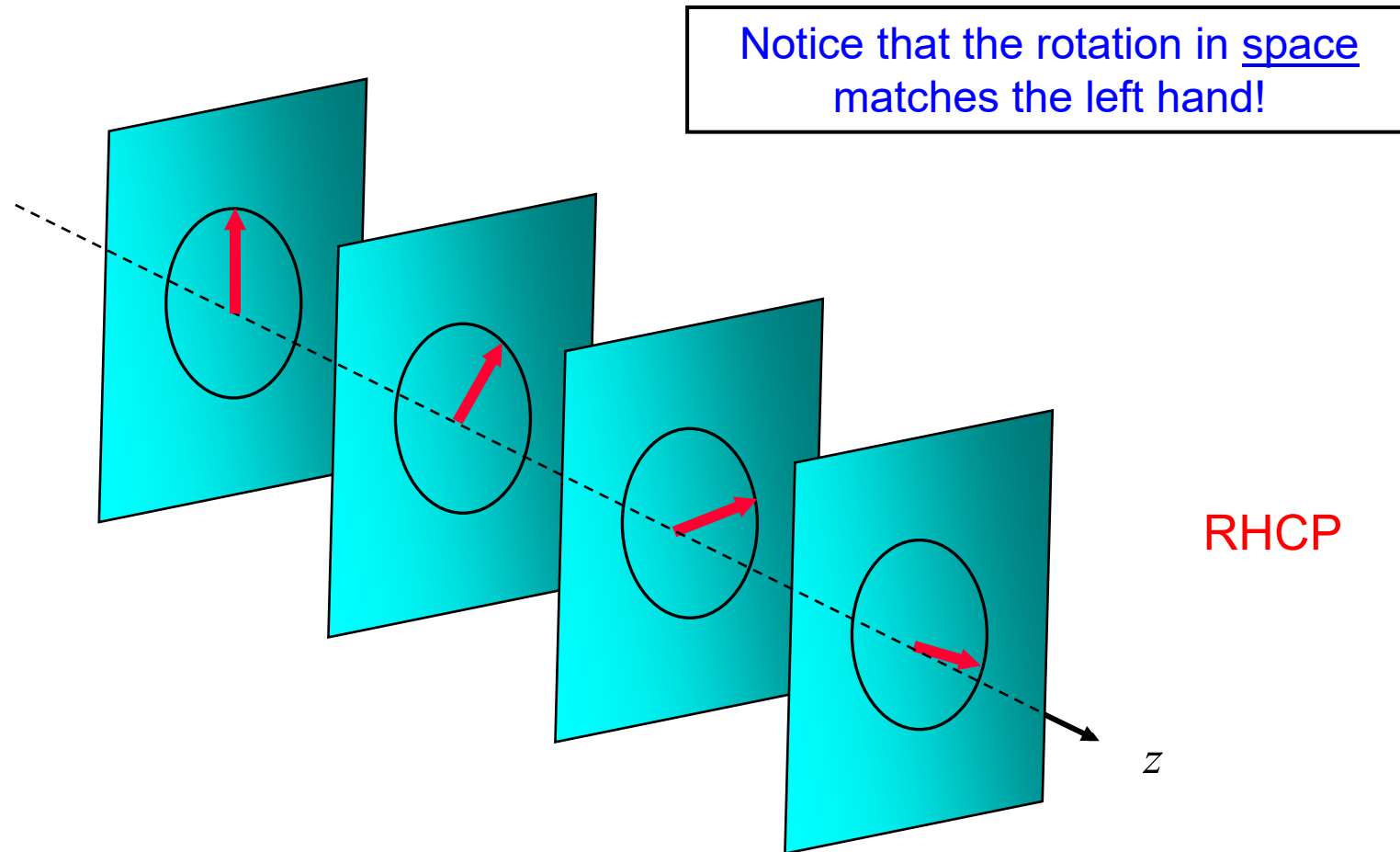
$$\underline{\mathcal{E}}(z, t) = \underline{\hat{x}} a \cos(\omega t - k_0 z) + \underline{\hat{y}} b \cos(\omega t - k_0 z + \beta) \quad \text{Time domain}$$



There is opposite rotation in space and time, due to the minus sign.

Circular Polarization (cont.)

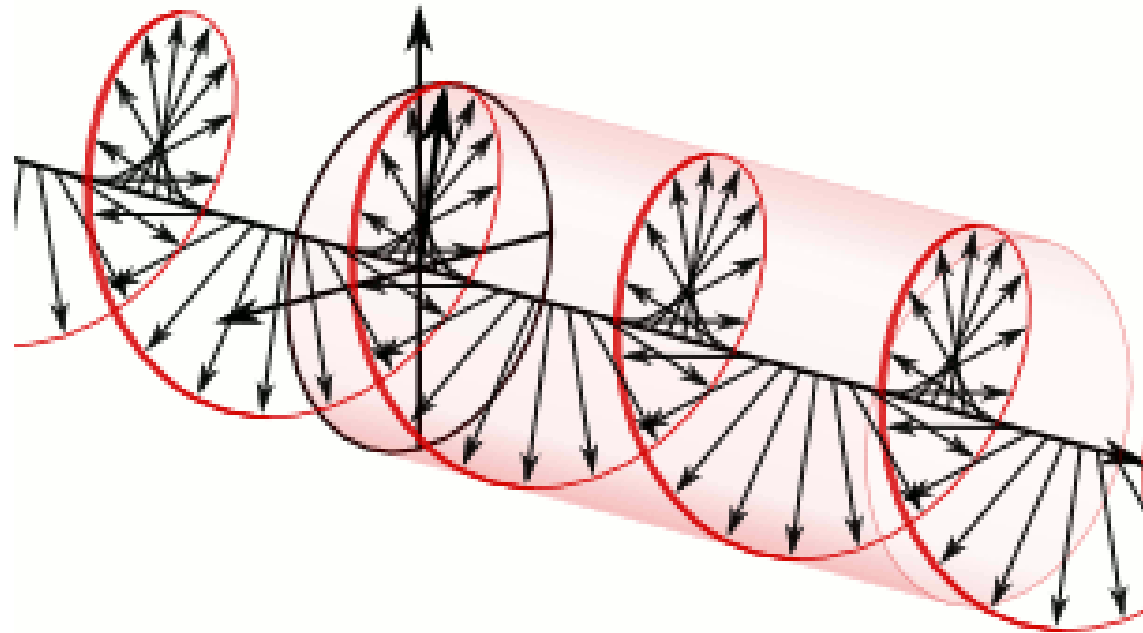
A snapshot of the electric field vector, showing the vector at different points.



Circular Polarization (cont.)

Animation of LHCP wave

(Use pptx version in full-screen mode to see motion.)

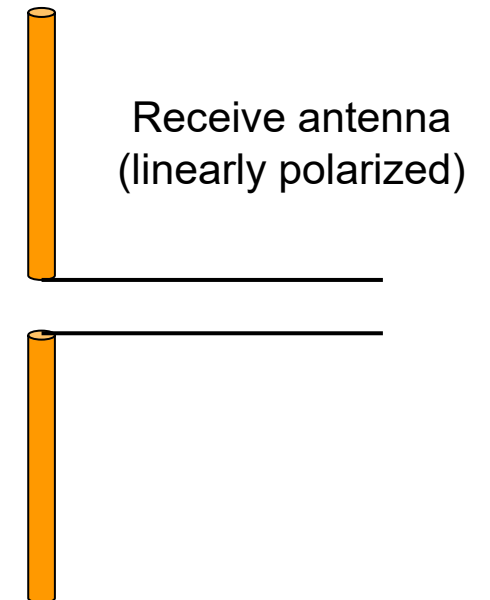
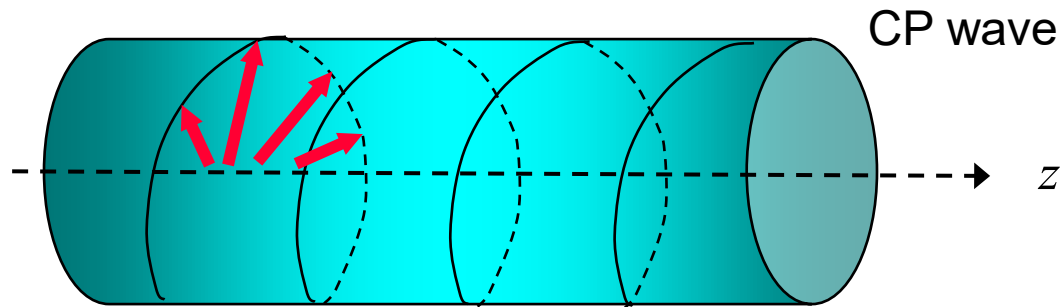


http://en.wikipedia.org/wiki/Circular_polarization

Circular Polarization (cont.)

Circular polarization is often used in wireless communications to avoid problems with signal loss due to polarization mismatch.

- Misalignment of transmit and receive antennas
- Reflections from buildings
- Propagation through the ionosphere



The linearly-polarized receive antenna will always receive a signal, no matter how it is rotated about the z axis.

However, for the same incident power density, an optimum linearly-polarized wave will give the maximum output signal from this linearly-polarized receive antenna (3 dB higher than from an incident CP wave). The linear receive antenna “throws away” half of the incident signal.

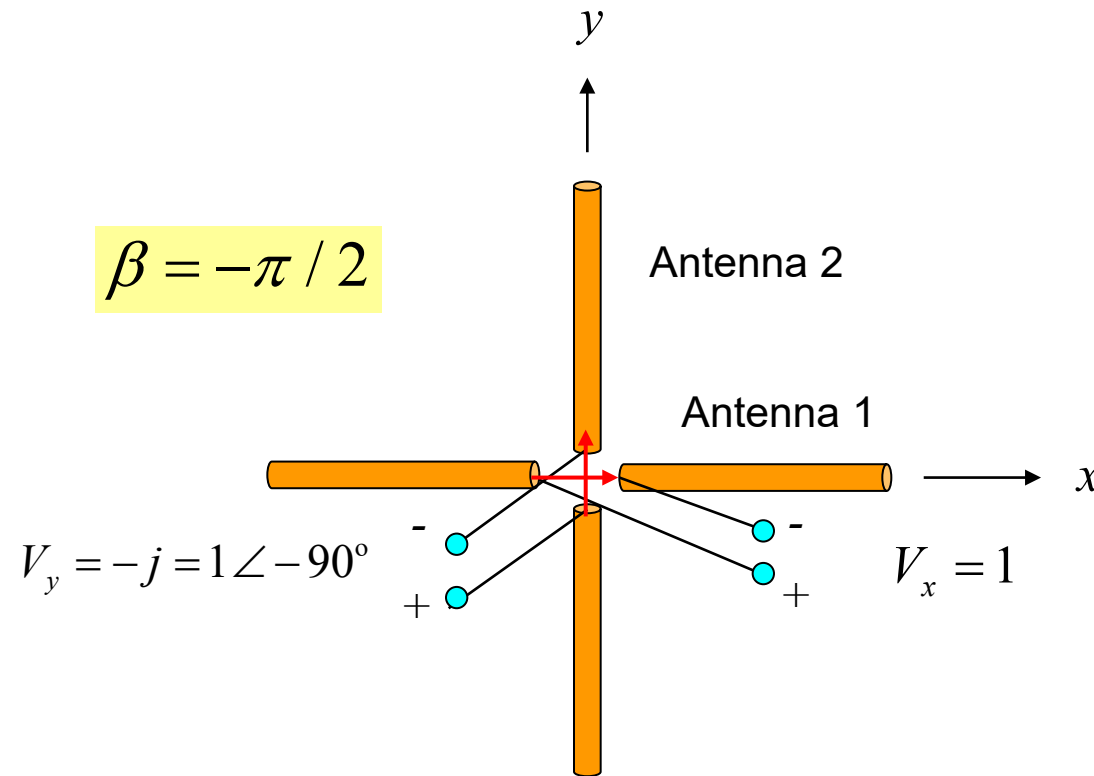
$$\underline{S} = \frac{1}{2\eta_0} \hat{z} \left(|E_x|^2 + |E_y|^2 \right)$$

Circular Polarization (cont.)

Two ways in which circular polarization can be obtained:

Method 1)

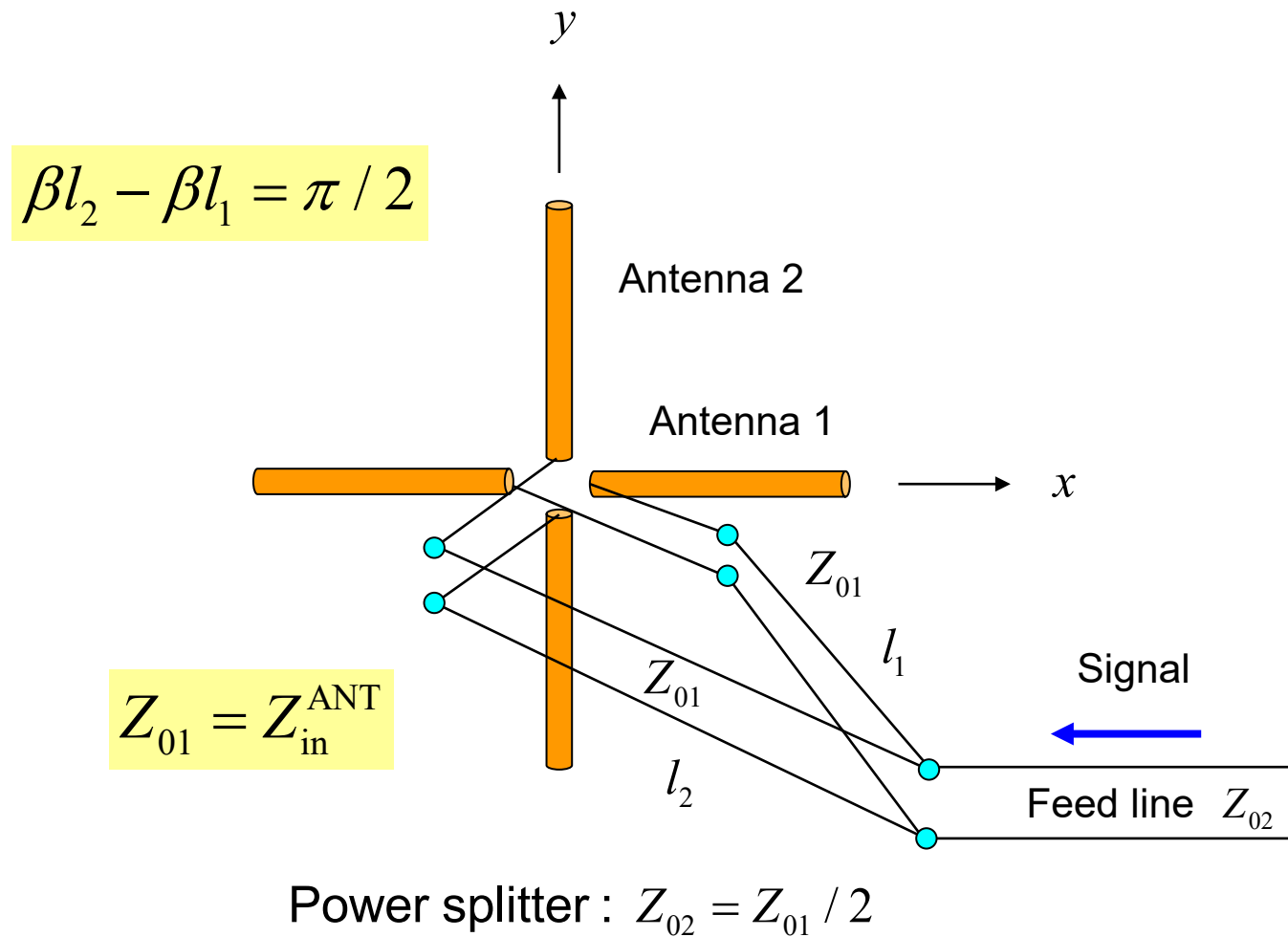
Use two identical antennas that are rotated by 90° , and also fed 90° out of phase.



This antenna will radiate a RHCP signal in the positive z direction, and LHCP in the negative z direction.

Circular Polarization (cont.)

Realization of method 1 using a 90° delay line:



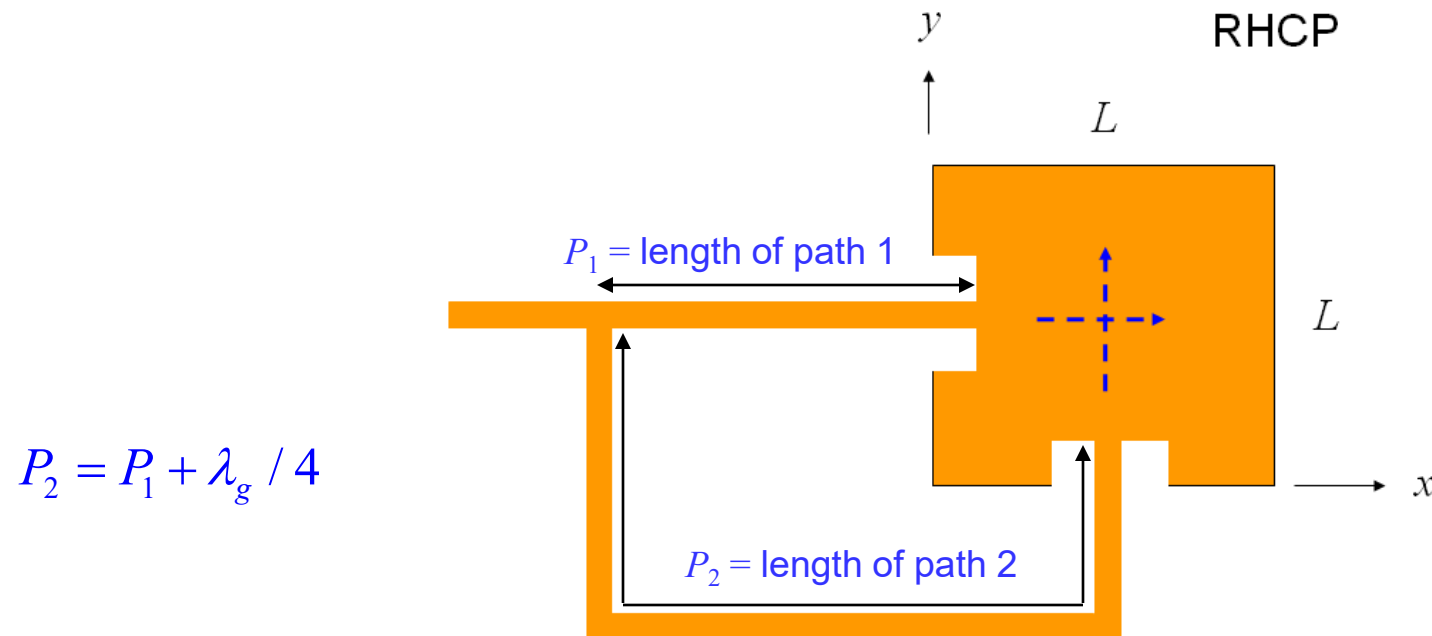
Circular Polarization (cont.)

An array of CP antennas



Circular Polarization (cont.)

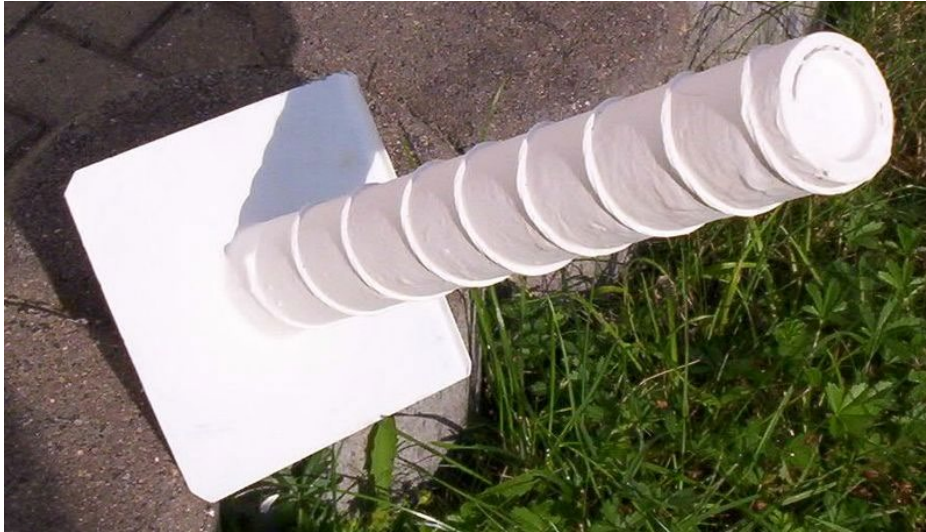
The two antennas can be realized by using two different modes of a single microstrip or dielectric resonator antenna.



RHCP for radiation in the $+z$ direction.

Circular Polarization (cont.)

Method 2) Use an antenna that inherently radiates circular polarization.



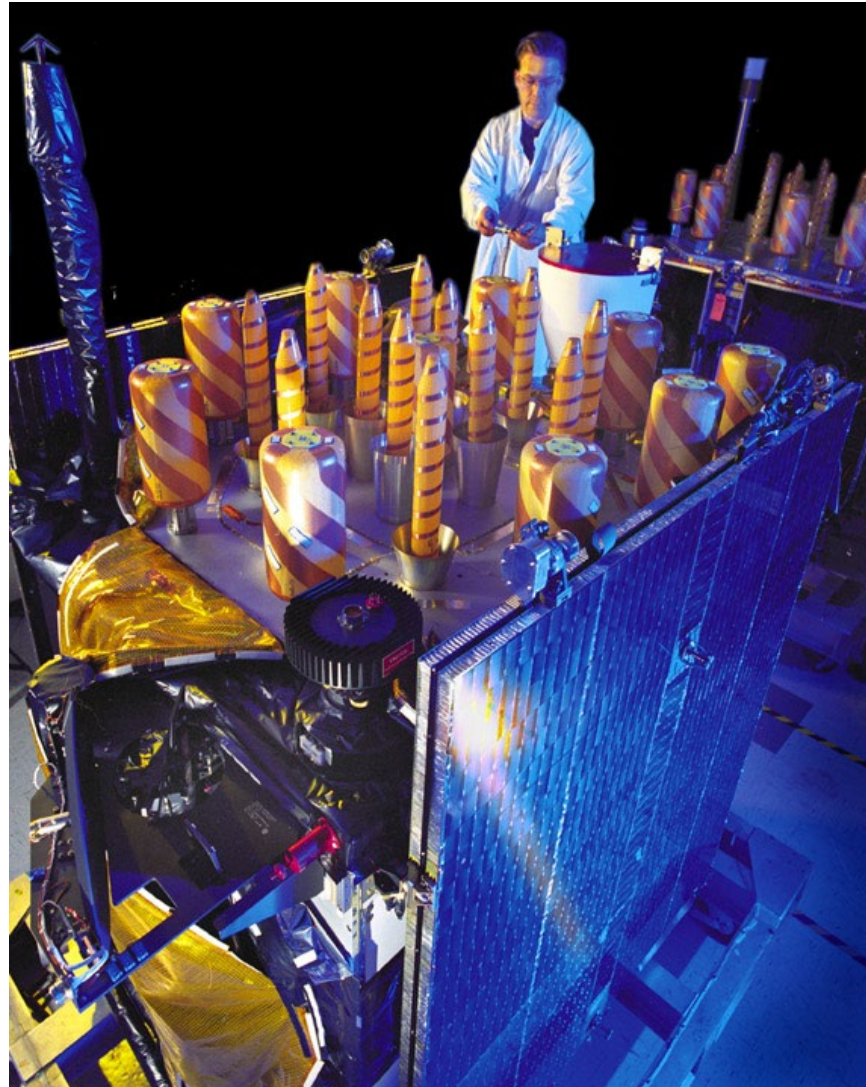
Helical antenna for WLAN communication at 2.4 GHz.



Array of four axial-mode helical antennas used as a satellite tracking-acquisition antenna, Pleumeur-Bodou, France.

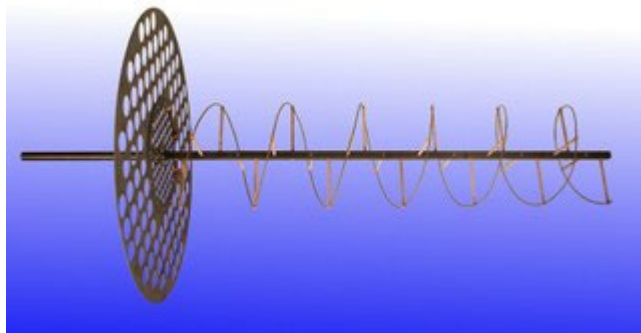
Circular Polarization (cont.)

Helical antennas on a GPS satellite



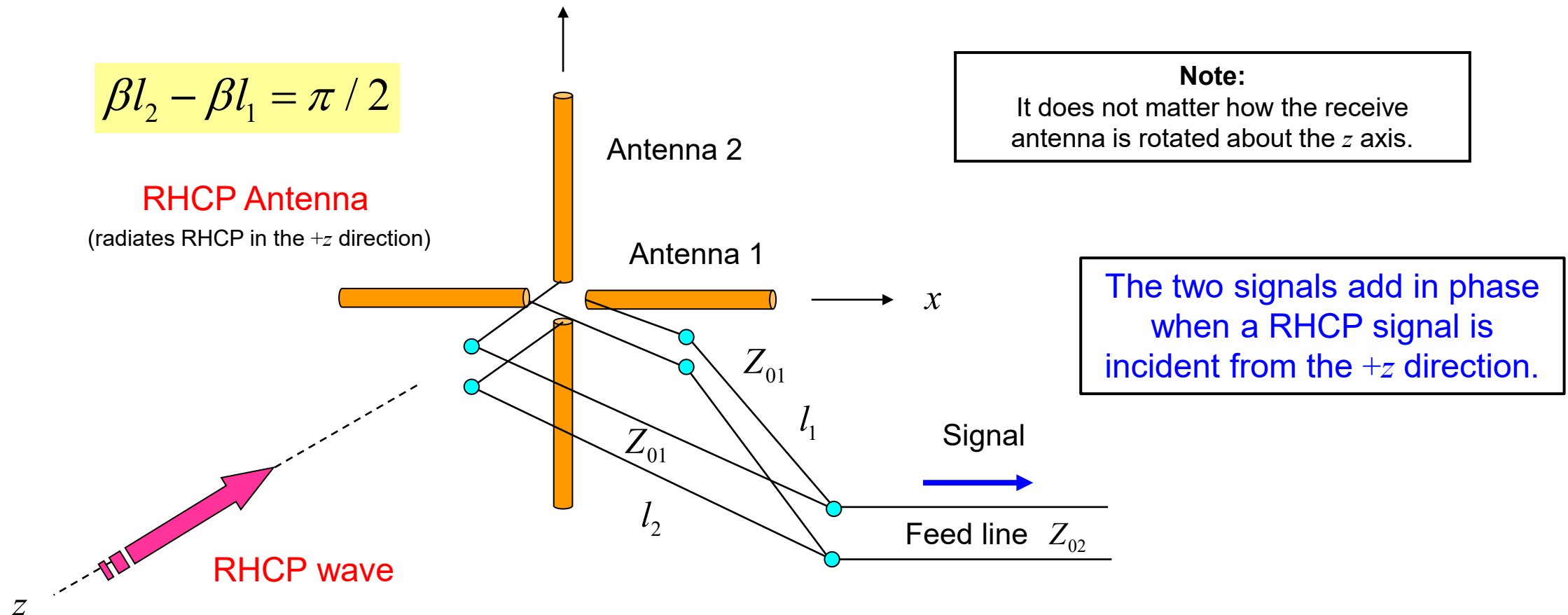
Circular Polarization (cont.)

Other Helical antennas



Circular Polarization (cont.)

An antenna that radiates circular polarization will also receive circular polarization of the same handedness, and be blind to the opposite handedness.
(The proof is omitted.)



Circular Polarization (cont.)

Summary of Possible Polarization Scenarios

1) Transmit antenna is LP, receive antenna is LP

- Simple, works well if both antennas are aligned.
- The received signal is less if there is a misalignment.

2) Transmit antenna is CP, receive antenna is LP

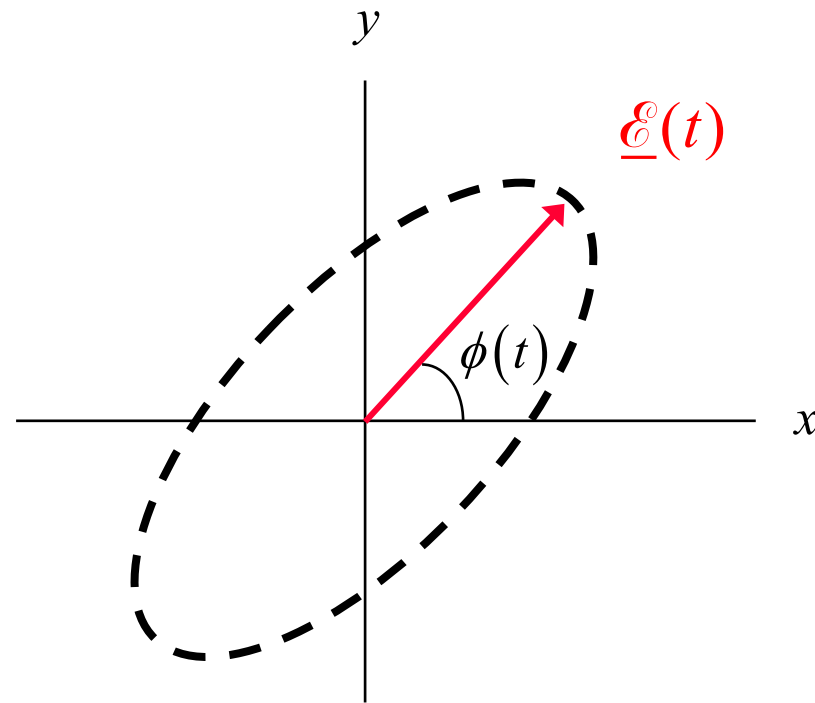
- Signal can be received no matter what the alignment of the receive antenna is.
- The received signal is 3 dB less than for two aligned LP antennas (for the same transmit power).

3) Transmit antenna is CP, receive antenna is CP (of the same handedness)

- Signal can be received no matter what the alignment of the receive antenna is.
- There is never a loss of signal, no matter what the alignment of the receive antenna is.
- The system is now more complicated.

Elliptic Polarization

Includes all other cases that are not linear or circular



The tip of the electric field vector stays on an ellipse.

(This is proved in Appendix A.)

Elliptic Polarization (cont.)

Rotation Property

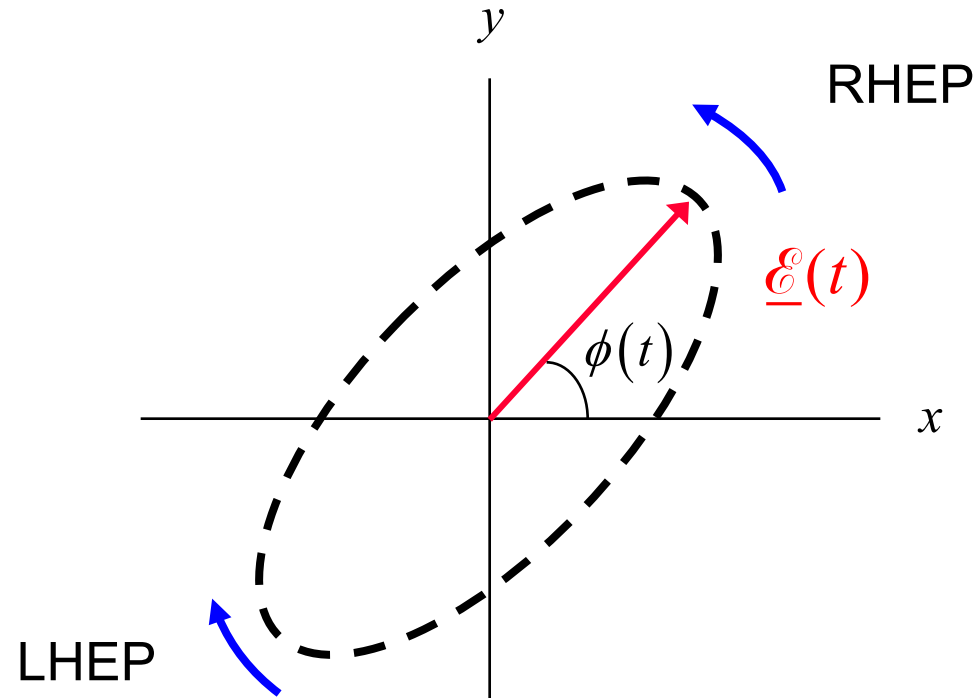
$$\mathcal{E}_x = a \cos(\omega t)$$

$$\mathcal{E}_y = b \cos(\omega t + \beta)$$

Recall:

$$E_x = a = \text{real number}$$

$$E_y = b e^{i\beta}$$



$$\begin{array}{ll} 0 < \beta < \pi & \text{LHEP} \\ -\pi < \beta < 0 & \text{RHEP} \end{array}$$

(This is proved in Appendix B.)

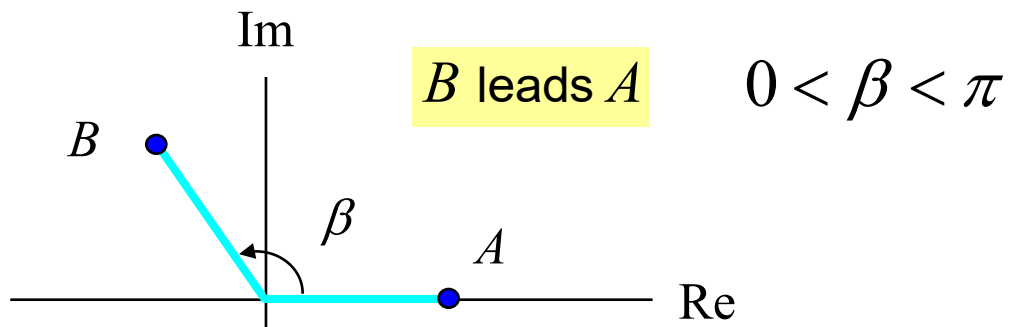
Rotation Rule

Here we give a simple graphical method for determining the type of polarization (left-handed or right-handed).

Rotation Rule (cont.)

First, we review the concept of leading and lagging sinusoidal waves.

Two phasors: A and B



$$A = ae^{j0}$$

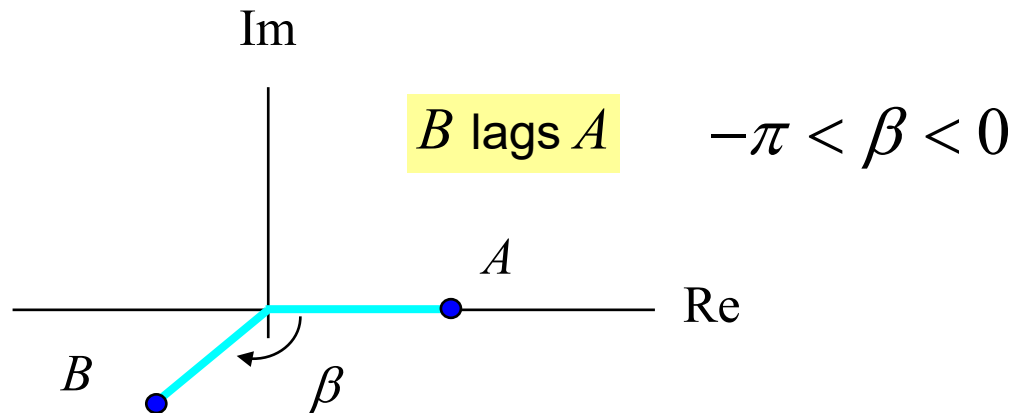
$$B = be^{j\beta}$$

$$\mathcal{A} = a \cos(\omega t)$$

$$\mathcal{B} = b \cos(\omega t + \beta)$$

Note:

We can always assume that the phasor A is on the real axis (zero degrees phase) without loss of generality, since it is only the phase difference between the two phasors that is important.



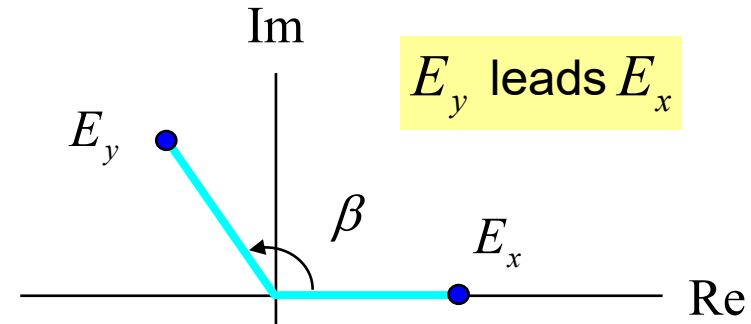
Note:

A lagging sinusoidal wave will appear shifted to the right (later time) on an oscilloscope trace.

Rotation Rule (cont.)

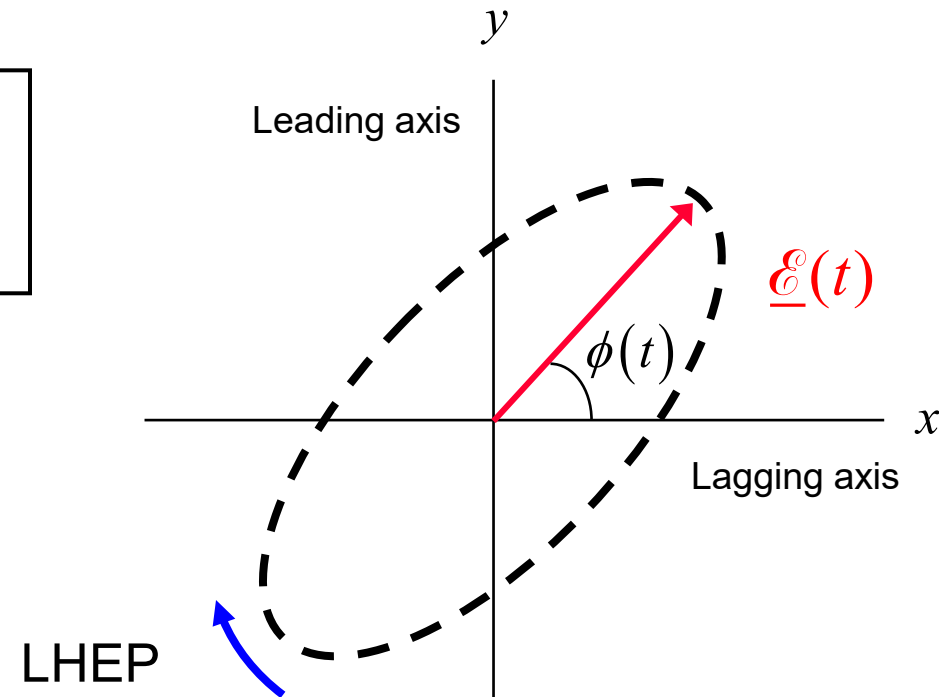
Now consider the case of a plane wave.

(a) $0 < \beta < \pi$ LHEP



Observation:

The electric field vector rotates in time from the leading axis to the lagging axis.

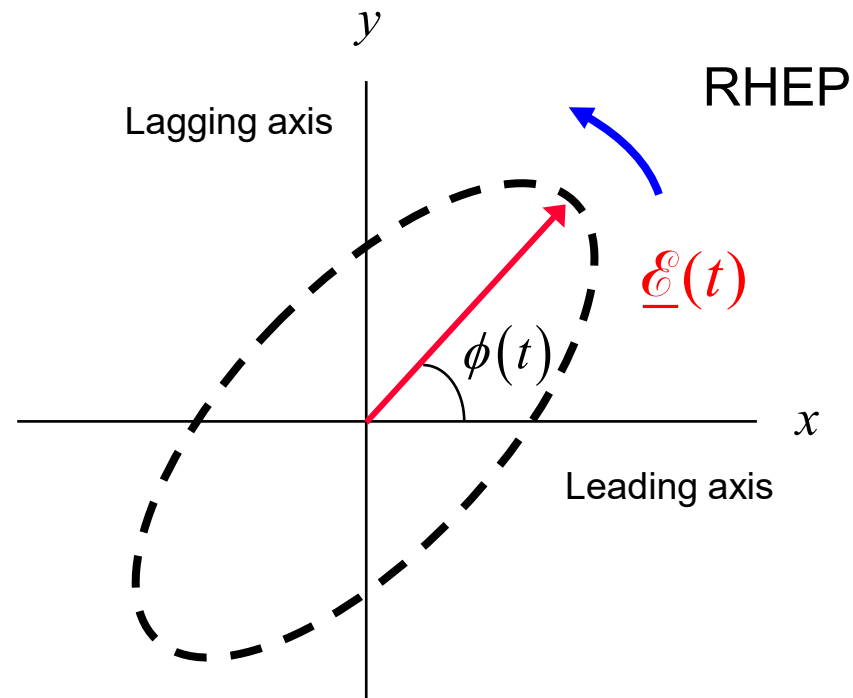
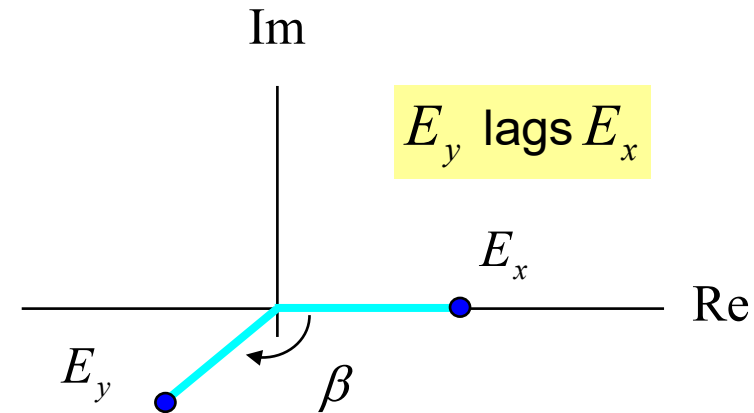


Rotation Rule (cont.)

(b) $-\pi < \beta < 0$ RHEP

Observation:

The electric field vector rotates in time from the leading axis to the lagging axis.



Rotation Rule (cont.)

The rule works in both cases, so we can call it a general rule:

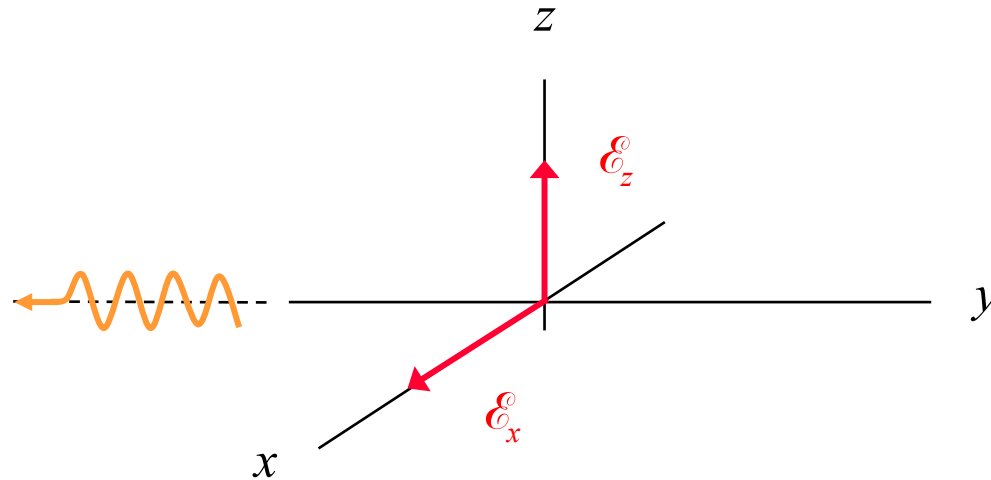
Rotation Rule:

In time, the electric field vector rotates from the leading axis to the lagging axis.

Rotation Rule (cont.)

Example

$$\underline{E} = [\underline{\hat{z}}(1 + j) + \underline{\hat{x}}(2 - j)]e^{jky}$$

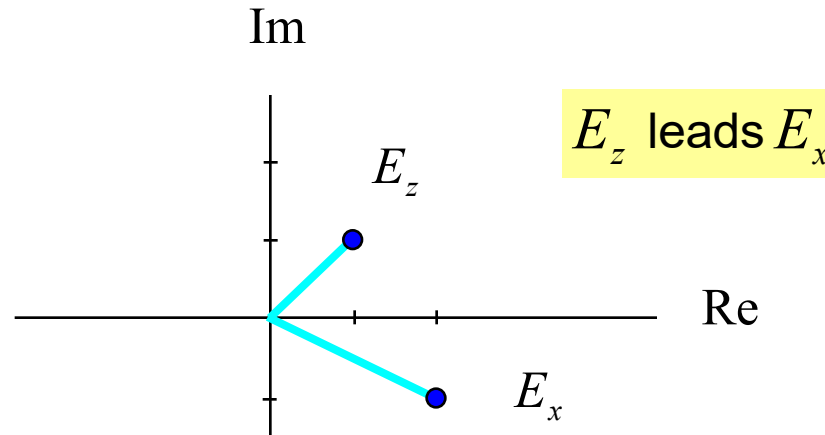


What is this wave's polarization?

Rotation Rule (cont.)

Example (cont.)

$$\underline{E} = [\underline{\hat{z}}(1 + j) + \underline{\hat{x}}(2 - j)]e^{jky}$$

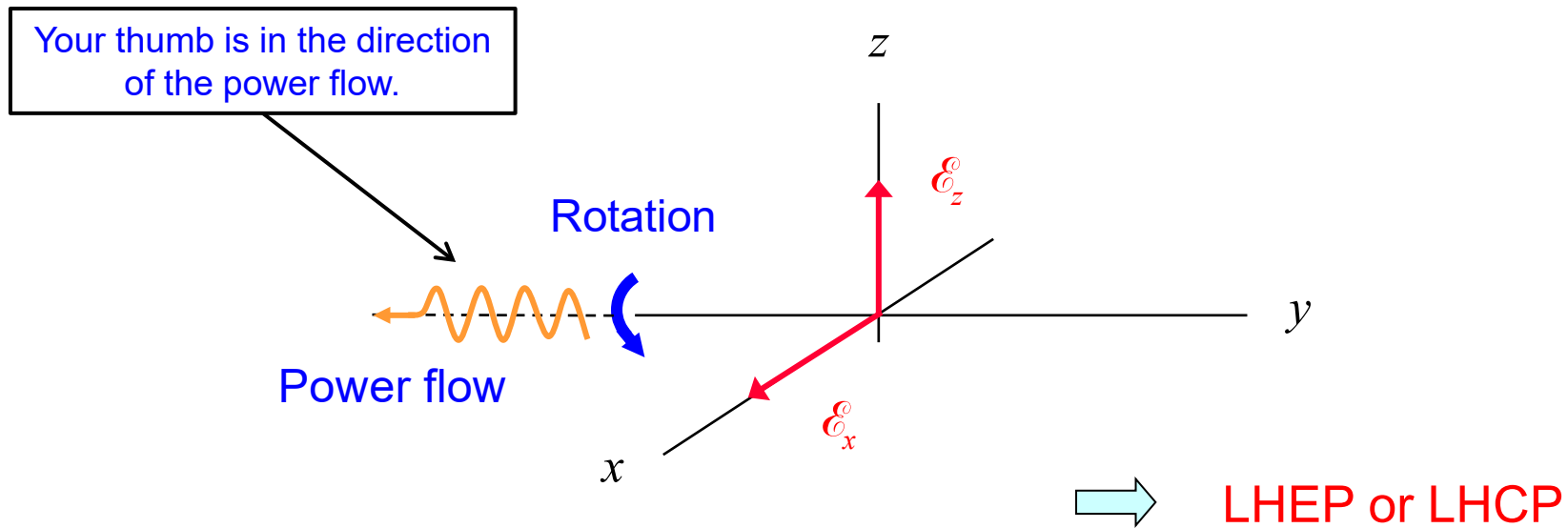


Therefore, in time the wave rotates from the z axis to the x axis.

Rotation Rule (cont.)

Example (cont.)

$$\underline{E} = [\underline{\hat{z}}(1+j) + \underline{\hat{x}}(2-j)]e^{jky}$$

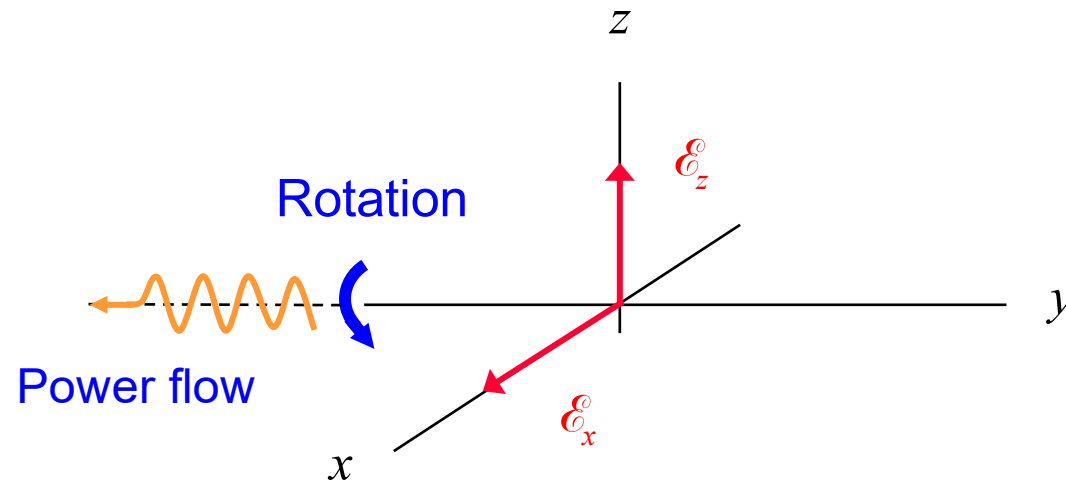


Note: $|E_x| \neq |E_z|$ and $\beta \neq \pm \frac{\pi}{2}$ (so this is not LHCP)

Rotation Rule (cont.)

Example (cont.)

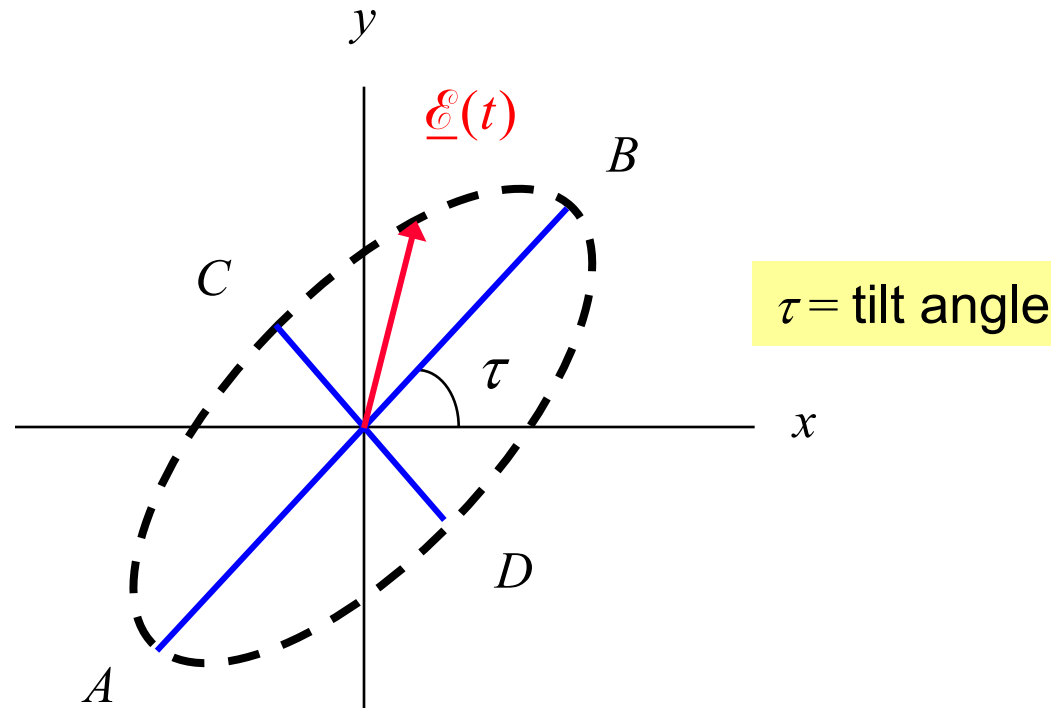
$$\underline{E} = [\underline{\hat{z}}(1 + j) + \underline{\hat{x}}(2 - j)]e^{jky}$$



Conclusion:

LHEP

Axial Ratio (AR) and Tilt Angle (τ)



$$\text{AR} \equiv \frac{\text{major axis}}{\text{minor axis}} = \frac{AB}{CD} > 1$$

Note: In dB we have $\text{AR}_{\text{dB}} = 20 \log_{10} (\text{AR})$

Axial Ratio (AR) and Tilt Angle (τ) Formulas

These formulas assume that the wave has E_x and E_y components, and the power is flowing in the $+z$ direction.

Recall:

$$E_x = a = \text{real number}$$

$$E_y = be^{j\beta}$$

$$\gamma \equiv \tan^{-1} \left(\frac{b}{a} \right)$$
$$0 \leq \gamma \leq 90^\circ$$

Tilt Angle

$$\tan 2\tau = \tan 2\gamma \cos \beta$$

Note:

The tilt angle τ is ambiguous by the addition of $\pm 90^\circ$.

Axial Ratio and Handedness

$$\text{AR} = |\cot \xi|$$

$$\xi > 0: \text{ LHEP}$$

$$\xi < 0: \text{ RHEP}$$

where

$$\sin 2\xi = \sin 2\gamma \sin \beta$$

$$-45^\circ \leq \xi \leq +45^\circ$$

Note on Tilt Angle

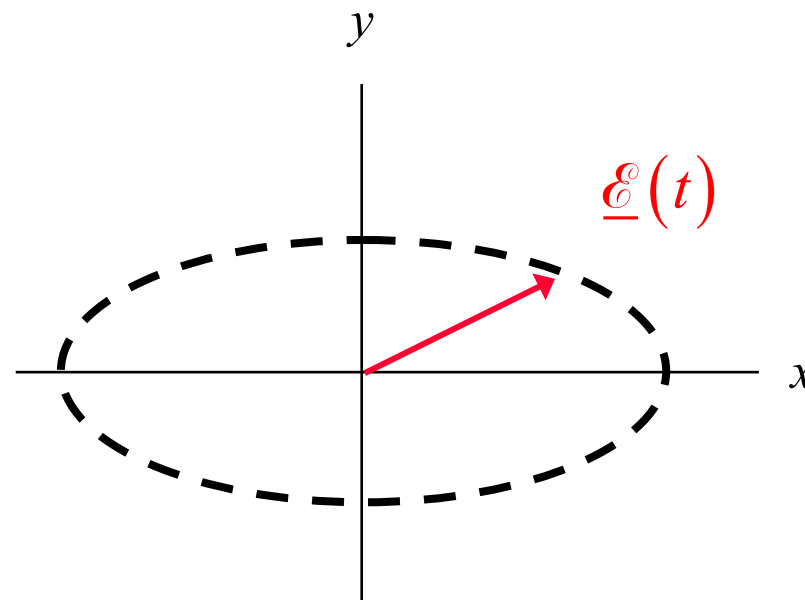
The title angle τ is zero or 90° if:

$$\beta = \pm\pi / 2$$

Tilt Angle: $\tan 2\tau = \tan 2\gamma \cos \beta$

$$\Rightarrow 2\tau = 0 \text{ or } 180^\circ$$

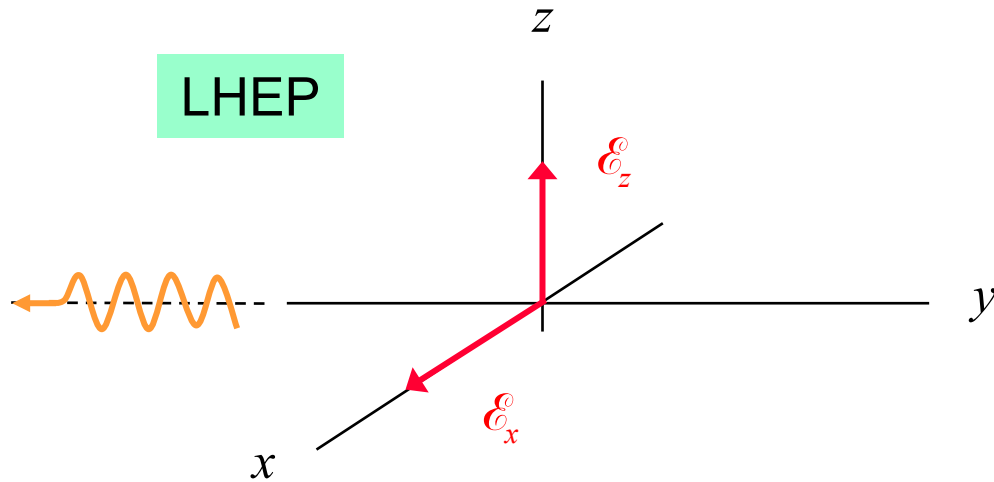
$$\Rightarrow \tau = 0 \text{ or } 90^\circ$$



Example

$$\underline{E} = [\underline{\hat{z}}(1+j) + \underline{\hat{x}}(2-j)]e^{jky}$$

Find the axial ratio and tilt angle.



Note:

In order to use the formulas for tilt angle and axial ratio, we need to relabel to coordinate system so that the wave has E_x and E_y components, and the power is flowing in the $+z$ direction.

Re-label the coordinate system:

Note: The new coordinate system needs to be a valid right-handed coordinate system:

$$\underline{\hat{x}} \times \underline{\hat{y}} = \underline{\hat{z}}$$

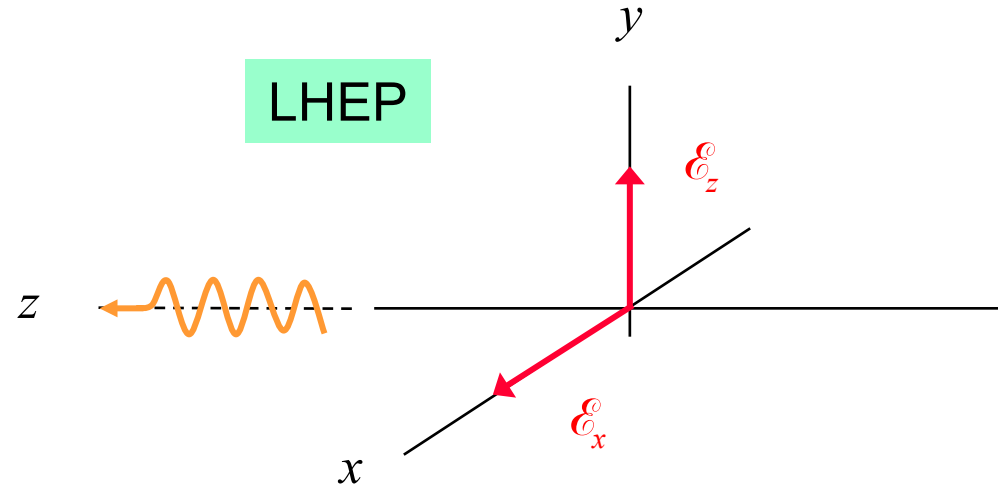
$$x \rightarrow x$$

$$z \rightarrow y$$

$$y \rightarrow -z$$

Example (cont.)

$$\underline{E} = \left[\underline{\hat{y}}(1+j) + \underline{\hat{x}}(2-j) \right] e^{-jkz}$$



Normalize: $\underline{E} = \left[\underline{\hat{x}}(1) + \underline{\hat{y}} \left(\frac{1+j}{2-j} \right) \right] e^{-jkz}$ or $\underline{E} = \left[\underline{\hat{x}}(1) + \underline{\hat{y}} (0.6324 e^{j1.249}) \right] e^{-jkz}$

Example (cont.)

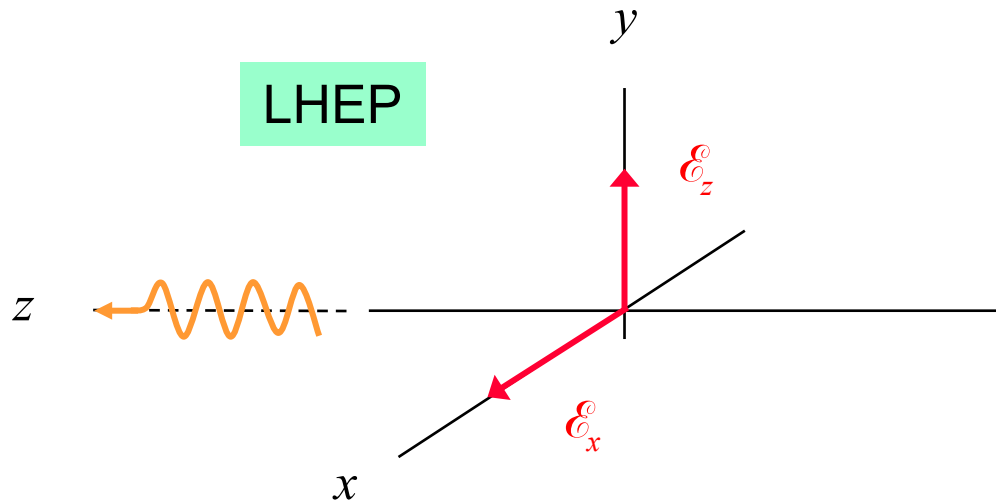
$$\underline{E} = \left[\underline{\hat{x}}(1) + \underline{\hat{y}}(0.6324 e^{j1.249}) \right] e^{-jkz}$$

Hence

$$a = 1$$

$$b = 0.6324$$

$$\beta = 1.249[\text{rad}] = 71.565^\circ$$



$$\beta = \angle E_y - \angle E_x = 71.565^\circ$$

$$\gamma = \tan^{-1} \left(\frac{b}{a} \right) = \tan^{-1} (0.632) = 0.564 [\text{rad}]$$

Example (cont.)

$$\tan 2\tau = \tan 2\gamma \cos \beta$$

$$\text{AR} = |\cot \xi|$$

$$\xi > 0: \text{ LHEP}$$

$$\xi < 0: \text{ RHEP}$$

where

$$\sin 2\xi = \sin 2\gamma \sin \beta$$

$$-45^\circ \leq \xi \leq +45^\circ$$

$$\gamma = 0.564 \text{ [rad]}$$

$$\beta = 71.565^\circ$$

Results:

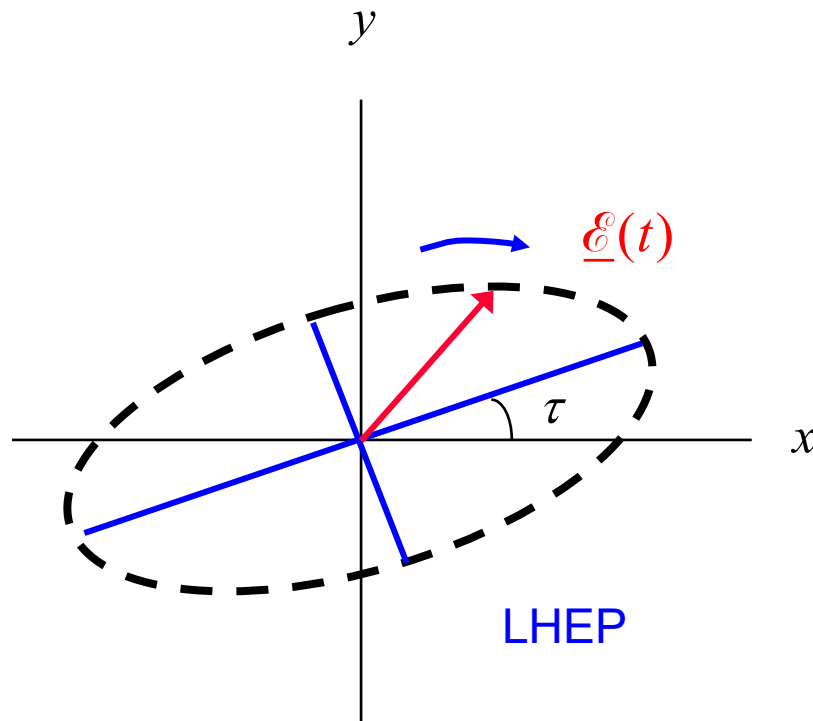
$$\tau = 16.845^\circ$$

$$\xi = 29.499^\circ$$

$$\text{AR} = 1.768$$

LHEP

Example (cont.)



$$AR = 1.768$$

$$\tau = 16.845^\circ$$

Note: We are not sure
which choice is correct:

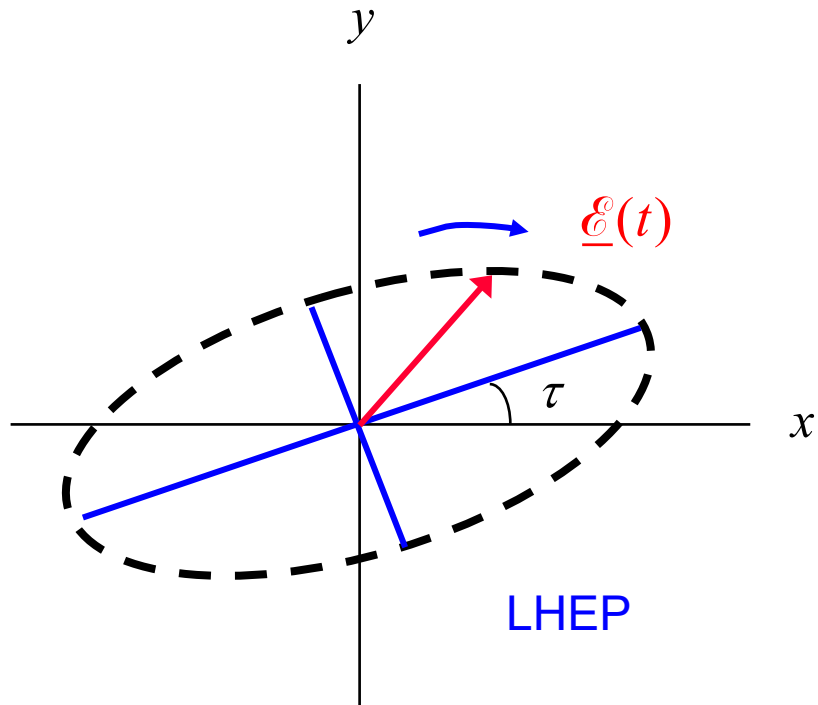
$$\tau = 16.845^\circ \quad \checkmark$$

$$\tau = 16.845^\circ + 90^\circ \quad \times$$

We can make a quick time-domain sketch to be sure.

Example (cont.)

Summary



$$\mathcal{E}_x = a \cos(\omega t)$$

$$\mathcal{E}_y = b \cos(\omega t + \beta)$$

Given:

$$a = 1$$

$$b = 0.6324$$

$$\beta = 1.249[\text{rad}] = 71.565^\circ$$

Results:

LHEP

$$\text{AR} = 1.768$$

$$\tau = 16.845^\circ$$

Appendix A

Here we give a proof that the tip of the electric field vector must stay on an ellipse.

$$\begin{aligned}\mathcal{E}_x &= a \cos \omega t & \mathcal{E}_y &= b \cos(\omega t + \beta) \\ & & &= b \cos \omega t \cos \beta - b \sin \omega t \sin \beta\end{aligned}$$

so

$$\mathcal{E}_y = b \cos \beta \left[\frac{\mathcal{E}_x}{a} \right] - b \sin \beta \left[\sqrt{1 - \left(\frac{\mathcal{E}_x}{a} \right)^2} \right] \quad \left(\sin \omega t = \sqrt{1 - \cos^2(\omega t)} \right)$$

or

$$\mathcal{E}_y - \mathcal{E}_x \left[\frac{b}{a} \cos \beta \right] = -\sin \beta \sqrt{b^2 - \left(\frac{b}{a} \right)^2 \mathcal{E}_x^2}$$

Squaring both sides, we have:

$$\mathcal{E}_y^2 + \mathcal{E}_x^2 \left[\frac{b}{a} \cos \beta \right]^2 - 2\mathcal{E}_x \mathcal{E}_y \left(\frac{b}{a} \cos \beta \right) = \sin^2 \beta \left[b^2 - \left(\frac{b}{a} \right)^2 \mathcal{E}_x^2 \right]$$

Appendix A (cont.)

$$\mathcal{E}_y^2 + \mathcal{E}_x^2 \left[\frac{b}{a} \cos \beta \right]^2 - 2\mathcal{E}_x \mathcal{E}_y \left(\frac{b}{a} \cos \beta \right) = \sin^2 \beta \left[b^2 - \left(\frac{b}{a} \right)^2 \mathcal{E}_x^2 \right]$$

Collecting terms, we have:

$$\mathcal{E}_x^2 \left[\left(\frac{b}{a} \right)^2 (\cos^2 \beta + \sin^2 \beta) \right] + \mathcal{E}_y^2 - 2\mathcal{E}_x \mathcal{E}_y \left(\frac{b}{a} \cos \beta \right) = b^2 \sin^2 \beta$$

or

$$\mathcal{E}_x^2 \left(\frac{b}{a} \right)^2 + \mathcal{E}_x \mathcal{E}_y \left[-2 \frac{b}{a} \cos \beta \right] + \mathcal{E}_y^2 = b^2 \sin^2 \beta$$

This is in the form of a quadratic expression:

$$A \mathcal{E}_x^2 + B \mathcal{E}_x \mathcal{E}_y + C \mathcal{E}_y^2 = D$$

Appendix A (cont.)

Discriminant:

$$\Delta = B^2 - 4AC \quad (\text{determines the type of curve})$$

$$= 4\left(\frac{b}{a}\right)^2 \cos^2 \beta - 4\left(\frac{b}{a}\right)^2$$

$$= 4\left(\frac{b}{a}\right)^2 [\cos^2 \beta - 1]$$

so

$$\Delta = -4\left(\frac{b}{a}\right)^2 \sin^2 \beta < 0$$

Hence, this is an ellipse.

(This follows from analytic geometry.)

Appendix B

Here we give a proof of the rotation property.

Rotation property:

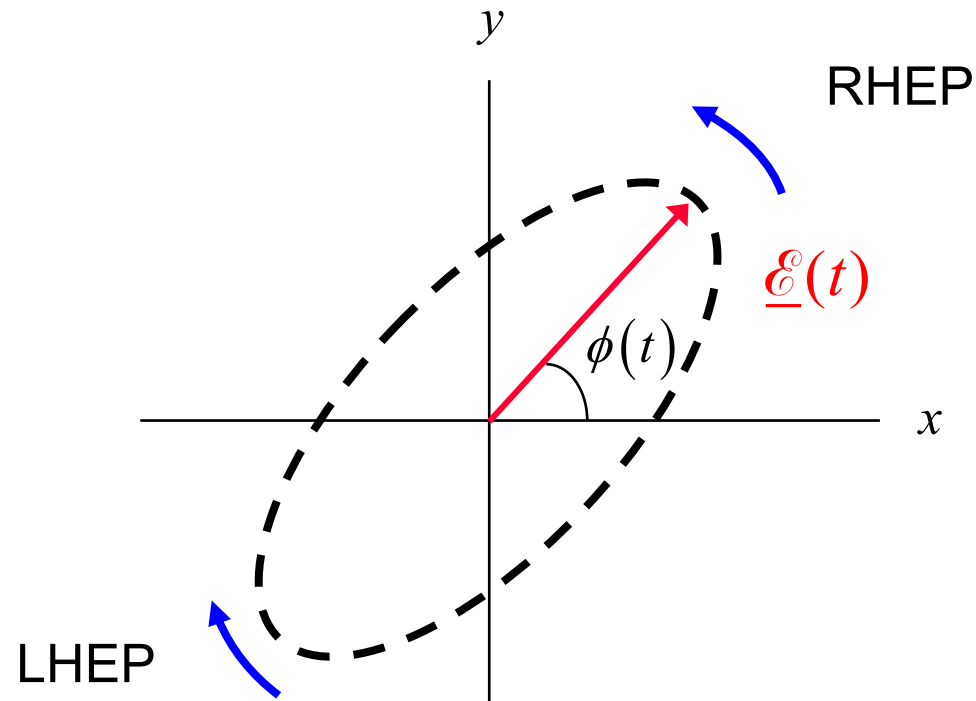
$$0 < \beta < \pi \quad \text{LHEP}$$

$$-\pi < \beta < 0 \quad \text{RHEP}$$

$$\mathcal{E}_x = a \cos \omega t$$

$$\mathcal{E}_y = b \cos(\omega t + \beta)$$

$$= b \cos \omega t \cos \beta - b \sin \omega t \sin \beta$$



Appendix B (cont.)

$$\tan \phi = \frac{\mathcal{E}_y}{\mathcal{E}_x} = \frac{b}{a} [\cos \beta - \tan \omega t \sin \beta]$$

$$\mathcal{E}_x = a \cos \omega t$$

$$\mathcal{E}_y = b \cos \omega t \cos \beta - b \sin \omega t \sin \beta$$

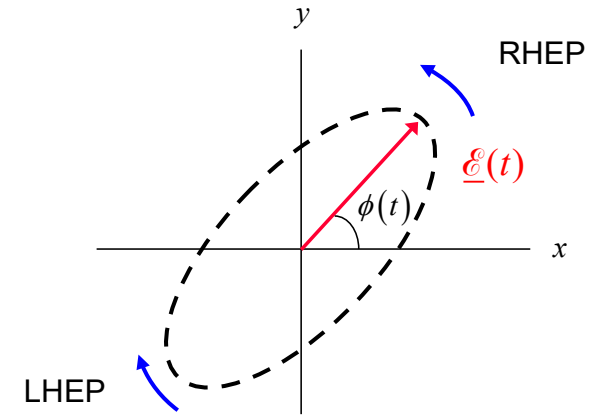
Take the derivative with respect to t :

$$\sec^2 \phi \frac{d\phi}{dt} = \left(\frac{b}{a} \right) [-\sec^2(\omega t)(\omega) \sin \beta]$$

Hence, we have:

$$\frac{d\phi}{dt} = -\sin \beta \left[\left(\frac{b}{a} \right) \cos^2 \phi \sec^2(\omega t)(\omega) \right]$$

The term in square brackets is always positive.



$$(a) \quad 0 < \beta < \pi \quad \frac{d\phi}{dt} < 0 \quad \text{LHEP}$$

$$(b) \quad -\pi < \beta < 0 \quad \frac{d\phi}{dt} > 0 \quad \text{RHEP} \quad (\text{proof complete})$$